

General Data	1	Tag Number					TBA																																											
	2	Service					TBA																																											
	3	PID No.					TBA																																											
	4	ITEM					DSH 6"300RF-INTEG TCV 1"300RF-HART																																											
	5	SIL Level Required		Line Number			NA			TBA																																								
Operating and Mechanical Data	6	STEAM CONDITIONS																																																
	7	INLET STEAM								OUTLET STEAM																																								
	8	Line Size		Piping Class		6"		A2A		6"		A2A																																						
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD																																						
	10	Insulation Code																																																
	11	Molecular Weight		Density		18.015				18.015																																								
	12	Flow :		Min.	Norm.	Max.	Unit	1500	2500	5000	KG/HR	*	*	*	KG/HR																																			
	13	Temp. :		Min.	Norm.	Max.	Unit	200	250	280	DEG C	150	165	170	DEG C																																			
	14	Press. :		Min.	Norm.	Max.	Unit	4	5	6	KG/CM2G	3.5	4	4.5	KG/CM2G																																			
	15	Temp. (after Desuperheater)		Flexibility																																														
	16	Design Press.		Design Temp. Max.		16.5				280																																								
	17	Body Material		Stem / Plug Material		A216 Gr.WCB				SS316																																								
	18	Size		Rating		Face		6"		300#		RF																																						
	19	Pipe Nozzle Length		Schedule		*				*																																								
	20	Insertion Length				*																																												
21	Flange Material				A105N																																													
Spray Water / Temperature Control Valve	22	WATER CONDITIONS																																																
	23	Line Size		Piping Class		1"				B2A																																								
	24	Line Material		Line Schedule		A106 Gr.B				XS																																								
	25	Insulation Code					Density																																											
	26	Flow :		Min.	Norm.	Max.	Unit	*	*	*		KG/HR																																						
	27	Temp. :		Min.	Norm.	Max.	Unit	105	105	110		DEG C																																						
	28	Press. :		Min.	Norm.	Max.	Unit	30	30	31		KG/CM2G																																						
	29	Design Temp.		Design Pressure		130				44																																								
	30	Body Material		Stem / Plug Material		A105N				SS316 (STELLITE)																																								
	31	Size		Rating		Face		1"		300		RF																																						
	32	Seat Leakage Class				IV																																												
	33																																																	
	34																																																	
	35	Area Classification Req		Safety Certification		IEC ZONE 1 IIC T3				Ex-ia																																								
	36	Instrument Type		Type																																														
	37	Packing Material		Bonnet Type		*				*																																								
	38	Bellows		Cooling Fins		Extension		NA		NA		NA																																						
	39	Air-Fail Position					OPEN																																											
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*																																							
41		Pneumatic Supply					3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)																																											
Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*																																								
	43	Fluid Connection		Electrical Connection		*				1/2"NPT																																								
	44	Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67																																								
	45	Surge Prot Device		Action Direction		YES				*																																								
Accessories	46	Handwheel		Position		YES				*																																								
	47	Locking Device		Filter		Press Reduc Valve		NA		*		*																																						
	48	Air Set		Pressure Gauge		Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA																																						
	49	Solenoid Valve		Tag No.		NA				NA																																								
Solenoid Valve	50	Elec Protect. Class		Enclosure Protection		NA																																												
	51	Position Detector		Tag No.		*				NA																																								
Position Detector	52	Elec Protect. Class		Enclosure Protection		NA				NA																																								
	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA																																								
Position Transmitter	54	Elec Protect. Class		Enclosure Protection		NA				NA																																								
	55			Weight		*				*																																								
Purchase	56	Manufacturer		Model No.		*				*																																								
	57	Client Reference		Requisition No.		NA				*																																								
	58	Material Code				TC9765132077																																												
Notes:																																																		
1) TBA: To be Assigned.																																																		
2) *: Vendor to Assign.																																																		
3) IBR Certification is applicable for Desuperheater & Temperature Control Valve.																																																		
4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST																																																		
5) Painting shall be as per attached Painting Specification																																																		
<table border="1"> <tr><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr><td></td><td></td><td></td><td></td><td></td><td></td></tr> <tr> <td>No.</td> <td>By</td> <td>Chk</td> <td>Appr</td> <td>Date</td> <td>Revision</td> </tr> </table> INSTRUMENT SPECIFICATION Desuperheater Code: 512 Dwg. No.: Ind Doc No.: Sheet Rev.: 																																													No.	By	Chk	Appr	Date	Revision
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General Data	1	Tag Number				TBA								
	2	Service				TBA								
	3	PID No.				TBA								
	4	ITEM				DSH 6"300RF-INTEG TCV 1"600RF-HART								
Operating and Mechanical Data	5	SIL Level Required		Line Number		NA				TBA				
	6					STEAM CONDITIONS								
	7					INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class		6"		A2A		6"		A2A		
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD		
	10	Insulation Code												
	11	Molecular Weight		Density		18.015				18.015				
	12	Flow :	Min.	Norm.	Max.	Unit	1500	2500	5000	KG/HR	*	*	*	KG/HR
	13	Temp. :	Min.	Norm.	Max.	Unit	200	250	280	DEG C	150	180	180	DEG C
	14	Press. :	Min.	Norm.	Max.	Unit	4	5	6	KG/CM2G	3	3.5	5	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility										
	16	Design Press.		Design Temp. Max.		16.5				320				
	17	Body Material		Stem / Plug Material		A216WCB				SS316				
	18	Size		Rating	Face	6"		300#		RF				
	19	Pipe Nozzle Length		Schedule		*				*				
	20	Insertion Length				*								
	21	Flange Material				A105N								
	22					WATER CONDITIONS								
	23	Line Size		Piping Class		1"				D2A				
	24	Line Material		Line Schedule		A106 Gr.B				XS				
	25	Insulation Code		Density										
	26	Flow :	Min.	Norm.	Max.	Unit	*		*		*		KG/HR	
	27	Temp. :	Min.	Norm.	Max.	Unit			105		170		DEG C	
	28	Press. :	Min.	Norm.	Max.	Unit			61		78		KG/CM2G	
	29	Design Temp.		Design Pressure		170				78.3				
	30	Body Material		Stem / Plug Material		A105N				SS316 (STELLITE)				
	31	Size		Rating	Face	1"		600		RF				
	32	Seat Leakage Class				IV								
	33													
	34													
	35	Area Classification		Req Safety Certification		IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type										
	37	Packing Material		Bonnet Type		*				*				
	38	Bellows	Cooling Fins	Extension		NA		NA		NA				
	39	Air-Fail Position				OPEN								
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*			
		41	Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)							
	Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*			
		43	Fluid Connection		Electrical Connection		*				1/2"NPT			
44		Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67				
Accessories	45	Surge Prot Device		Action Direction		YES				*				
	46	Handwheel		Position		YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*				
Solenoid Valve	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA				
	49	Solenoid Valve		Tag No.		NA				NA				
	50	Elec Protect. Class		Enclosure Protection		NA								
Position Detector	51	Position Detector		Tag No.		*				NA				
	52	Elec Protect. Class		Enclosure Protection		NA				NA				
Position Transmitter	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA				
	54	Elec Protect. Class		Enclosure Protection		NA				NA				
Purchase	55			Weight						*				
	56	Manufacturer		Model No.		*				*				
	57	Client Reference		Requisition No.		NA				*				
	58	Material Code				TC9765132581								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST														
						INSTRUMENT SPECIFICATION Desuperheater								
						Code: 512				Dwg. No.: Ind Doc No.:				
						Sheet				Rev.:				
No.	By	Chk	Appr	Date	Revision									

General Data	1	Tag Number				TBA								
	2	Service				TBA								
	3	PID No.				TBA								
	4	ITEM				DSH 4"300RF-NON INTEG TCV 1"600RF-HART								
	5	SIL Level Required		Line Number		NA				TBA				
Operating and Mechanical Data	6					STEAM CONDITIONS								
	7					INLET STEAM				OUTLET STEAM				
	8	Line Size		Piping Class		4"		A2A		4"		A2A		
	9	Line Material		Line Schedule		A106 Gr.B		STD		A106 Gr.B		STD		
	10	Insulation Code												
	11	Molecular Weight		Density		18.015				18.015				
	12	Flow :	Min.	Norm.	Max.	Unit	400	500	750	KG/HR	*	*	*	KG/HR
	13	Temp. :	Min.	Norm.	Max.	Unit	300	300	343	DEG C	150	180	180	DEG C
	14	Press. :	Min.	Norm.	Max.	Unit	5	5	6.5	KG/CM2G	3	3.5	5	KG/CM2G
	15	Temp. (after Desuperheater)		Flexibility										
	16	Design Press.		Design Temp. Max.		16.5				320				
	17	Body Material		Stem / Plug Material		A216WCB				SS316				
	18	Size		Rating	Face	4"		300#		RF				
	19	Pipe Nozzle Length		Schedule		*				*				
	20	Insertion Length				*								
	21	Flange Material				A105N								
	22					WATER CONDITIONS								
	23	Line Size		Piping Class		1"				D2A				
	24	Line Material		Line Schedule		A106 Gr.B				XS				
	25	Insulation Code		Density										
	26	Flow :	Min.	Norm.	Max.	Unit	*		*		*		KG/HR	
	27	Temp. :	Min.	Norm.	Max.	Unit			105		170		DEG C	
	28	Press. :	Min.	Norm.	Max.	Unit			61		78		KG/CM2G	
	29	Design Temp.		Design Pressure		170				78.3				
	30	Body Material		Stem / Plug Material		A216WCB				SS316 (STELLITE)				
	31	Size		Rating	Face	1"		600		RF				
	32	Seat Leakage Class				IV								
	33													
	34													
	35	Area Classification		Req Safety Certification		IEC ZONE 1 IIC T3				Ex-ia				
	36	Instrument Type		Type										
	37	Packing Material		Bonnet Type		*				*				
	38	Bellows	Cooling Fins	Extension		NA		NA		NA				
	39	Air-Fail Position				OPEN								
	Actuator	40	Actuator Type		Spring Range		SPRING & DIAPHRAGM				*			
		41	Pneumatic Supply				3.5 KG/CM2G (MIN), 4.0 KG/CM2G (NOR)							
	Positioner	42	Type		Input Signal		4-20MA SMART HART 7				*			
		43	Fluid Connection		Electrical Connection		*				1/2"NPT			
		44	Elec Protect. Class		Enclosure Protection		Ex-ia				IP 67			
	45	Surge Prot Device		Action Direction		YES				*				
Accessories	46	Handwheel		Position		YES				*				
	47	Locking Device	Filter	Press Reduc Valve		NA		*		*				
Solenoid Valve	48	Air Set	Pressure Gauge	Air Volume Tank		YES, SS316, 5 MICRON SS		50MM		NA				
	49	Solenoid Valve		Tag No.		NA				NA				
Position Detector	50	Elec Protect. Class		Enclosure Protection		NA								
	51	Position Detector		Tag No.		*				NA				
Position Transmitter	52	Elec Protect. Class		Enclosure Protection		NA				NA				
	53	Position Transmitter		Tag No.		INCLUDED WITH POSITIONER				NA				
Purchase	54	Elec Protect. Class		Enclosure Protection		NA				NA				
	55			Weight						*				
	56	Manufacturer		Model No.		*				*				
	57	Client Reference		Requisition No.		NA				*				
	58	Material Code				TC9765132603								
Notes: 1) TBA: To be Assigned. 2) *: Vendor to Assign. 3) IBR Certification is applicable for Desuperheater & Temperature Control Valve. 4) Following tests required: - MATERIAL TEST CERTIFICATE FOR TRIM AND BODY AS PER EN 3.1, - IGC-E FOR TRIM, - RADIOGRAPHY FOR CASTING, - DP/MP FOR CASTNG, - PMI FOR ASSEMBLED VALVES IF SS BODY, - HYDROTEST														
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